Area Graph (Area Plot)

An **Area Graph** is similar to a **line graph**, but the area **below the line is filled with color**. It shows **change over time** and also emphasizes the **magnitude of values**.

When to Use?

- Time-series data
- Trend + volume visualization
- Comparing growth over time

Basic Syntax

```
plt.fill_between(x, y)
```

Simple Example

```
import matplotlib.pyplot as plt

x = [1, 2, 3, 4, 5]
y = [10, 20, 15, 30, 25]

plt.fill_between(x, y)
plt.show()
```

Most Used Area Plot Parameters

Parameter	Description
`x, y`	Data values
`color`	Area color
`alpha`	Transparency
`label`	Legend label
`where`	Conditional filling
`interpolate`	Smooth edges

Example with Parameters

```
plt.fill_between(
    x, y,
    color='skyblue',
    alpha=0.5,
    label='Sales'
)
plt.legend()
plt.show()
```

MULTI-AREA GRAPH

What is a Multi-Area Graph?

A **Multi-Area Graph** displays **multiple area plots on the same axes** to compare trends between datasets.

Areas may **overlap**.

When to Use?

- Comparing multiple trends
- Same time scale
- Overlapping datasets

Example

```
y1 = [10, 20, 30, 40, 50]
y2 = [5, 15, 25, 35, 45]

plt.fill_between(x, y1, alpha=0.5, label='Product A')
plt.fill_between(x, y2, alpha=0.5, label='Product B')

plt.legend()
plt.show()
```

Important Parameters

Same as area plot +:

- `alpha` (very important to see overlap)
- `label` for legend

STACKED AREA GRAPH (STACK GRAPH)

A **Stacked Area Graph** shows **multiple datasets stacked on top of each other**, representing both:

- Individual contribution
- Total value

When to Use?

- Showing **part-to-whole relationship**
- Trend + composition over time
- Cumulative data

Basic Syntax

```
plt.stackplot(x, y1, y2, y3)
```

Simple Example

```
import matplotlib.pyplot as plt

x = [1, 2, 3, 4, 5]
y1 = [10, 20, 30, 40, 50]
y2 = [5, 15, 25, 35, 45]
y3 = [2, 10, 15, 20, 30]

plt.stackplot(x, y1, y2, y3)
plt.show()
```

Most Used Stack Plot Parameters

Parameter	Description
`x`	X-axis values
`y`	Multiple datasets
`labels`	Area names
`colors`	Area colors
`alpha`	Transparency
`baseline`	Stack reference

Example with Parameters

```
plt.stackplot(
    x,
    y1, y2, y3,
    labels=['A', 'B', 'C'],
    alpha=0.7
)
plt.legend()
plt.show()
```

Comparison Table

Plot Type	Use Case
Area Plot	Single trend
Multi-Area Plot	Compare trends
Stacked Area	Part-to-whole + trend

What is a Step Plot?

A **Step Plot** is a type of plot where data is drawn as a series of **horizontal and vertical lines**, forming steps.
The value remains **constant for an interval** and changes **suddenly** at specific points.

It looks like a **staircase**.

When to Use a Step Plot?

- Discrete or interval-based data
- Sudden value changes
- Digital signals (ON/OFF)
- Inventory, price changes, attendance counts

Basic Syntax

```
plt.step(x, y)
```

Simple Step Plot Example

```
import matplotlib.pyplot as plt

x = [1, 2, 3, 4, 5]
y = [10, 10, 20, 20, 30]

plt.step(x, y)
```

```
plt.xlabel("X Values")
plt.ylabel("Y Values")
plt.title("Basic Step Plot")
plt.show()
```

Most Used Step Plot Parameters

1. `where`

Controls **where the step change occurs**.

```
plt.step(x, y, where='mid')
```

Values:

- `'pre'` → step before x
- `'post'` → step after x (**default**)
- `'mid'` → step in the middle

2. `color`

Changes line color.

```
plt.step(x, y, color='green')
```

3. `linestyle`

Controls line style.

```
plt.step(x, y, linestyle='--')
```

4. `linewidth`

Controls thickness of the step line.

```
plt.step(x, y, linewidth=2)
```

5. `marker`

Adds markers at step points.

```
plt.step(x, y, marker='o')
```

6. `label`

Adds legend label.

```
plt.step(x, y, label='Data')
plt.legend()
```

Step Plot with Parameters Example

```
plt.step(
    x, y,
    where='post',
    color='blue',
```

```
    linestyle='--',
    linewidth=2,
    marker='o',
    label='Step Data'
)

plt.legend()
plt.show()
```

What is a Multi-Step Graph?

A **Multi-Step Graph** displays **multiple step plots on the same axes** to compare several datasets.

Useful for **comparison of discrete changes**.

When to Use?

- Comparing multiple digital signals
- Comparing step-wise trends
- Multiple categories changing over time

Multi-Step Graph Example

```
x = [1, 2, 3, 4, 5]

y1 = [5, 5, 10, 10, 15]
y2 = [3, 6, 6, 9, 12]

plt.step(x, y1, where='post', label='Product A')
plt.step(x, y2, where='post', label='Product B')

plt.xlabel("Time")
plt.ylabel("Value")
plt.title("Multi-Step Graph")
plt.legend()
plt.show()
```

Important Parameters for Multi-Step Graph

- `where` (very important)
- `label` (for legend)
- `color`, `linestyle`, `linewidth` (to differentiate lines)

Step Plot vs Line Plot

Step Plot	Line Plot
Sudden change	Smooth change
Discrete data	Continuous data
Constant intervals	Continuous variation